

VIP Pass Scanner — Firebase Setup Guide

This guide explains how to configure your own Firebase project for the VIPPassScanner Android application.

1. Create Your Own Firebase Project

Each person using the GitHub project should create and use their own Firebase project and database.

In the Firebase Console:

1. Create a new Firebase project.
2. Add an Android app.
3. Use this package name:
com.nikhil.vippassscanner
4. Register the app.
5. Download the generated google-services.json.

Place the downloaded file at:

```
app/google-services.json
```

2. Firebase Authentication

The application uses Firebase Authentication for staff login.

```
Firebase Console → Authentication → Sign-in method → Email/Password → Enable
```

Create at least one user account for testing/staff login.

3. Cloud Firestore

Open Firestore Database in the Firebase Console and create a Cloud Firestore database.

The application expects these collections:

```
passes
scanHistory
```

4. passes Collection

Each VIP pass is stored as a document. The document ID must match the QR/barcode value that will be scanned.

```
passes/VIP001
```

Field	Type	Description
used	Boolean	Whether the pass has already been used
scannedBy	String	Email of the staff member who scanned it
scannedAt	Timestamp	Time when the pass was scanned

For initial testing, a pass can be created as:

```
Collection: passes
Document ID: VIP001
```

```
used: false
```

If the QR code contains VIP001, the Firestore document should be passes/VIP001.

5. scanHistory Collection

The application creates scan-history records automatically after a valid pass is scanned. You do not need to manually create every history document.

```
scanHistory
■■■ automatically-generated-document-id
    ■■■ passCode: "VIP001"
    ■■■ status: "VALID"
    ■■■ scannedBy: "staff@example.com"
    ■■■ scannedAt: Timestamp
```

Field	Type	Description
passCode	String	Scanned pass code
status	String	Scan result
scannedBy	String	Staff member's email
scannedAt	Timestamp	Time of scan

6. How Pass Validation Works

```
QR/barcode value
↓
Find matching document in passes
↓
Does document exist?
    ■■■ No → INVALID PASS
    ■■■ Yes
        ↓
        Is used == true?
            ■■■ Yes → PASS ALREADY USED
            ■■■ No
                ↓
                Mark pass as used
                Store scannedBy and scannedAt
                Create scanHistory record
                ↓
                ACCESS GRANTED
```

7. Firestore Security Rules

For real deployment, Firestore should not be left open to everyone. A basic authenticated-user starting point is:

```
rules_version = '2';

service cloud.firestore {
  match /databases/{database}/documents {

    match /passes/{passId} {
      allow read, update: if request.auth != null;
    }

    match /scanHistory/{historyId} {
      allow read, create: if request.auth != null;
    }
  }
}
```

Review and customize the rules before using the application at a real event.

8. Testing

1. Create a Firestore document:
Collection: passes
Document ID: VIP001
used: false
2. Create a QR code containing:
VIP001
3. Log into the application.
4. Scan the QR code.

Expected result:
ACCESS GRANTED

5. The pass becomes used and a scanHistory record is created.
6. Scan VIP001 again.

Expected result:
PASS ALREADY USED

9. google-services.json

The repository intentionally excludes the real Firebase configuration. The project can contain a template named google-services.json.example. Each user should download their own google-services.json from their Firebase project.

```
app/google-services.json      ← your own Firebase file
app/google-services.json.example ← safe template for GitHub
```

10. AI Help Prompt

If someone needs help configuring Firebase, they can paste the following prompt into ChatGPT or another AI assistant:

I am setting up the Firebase backend for an Android project called VIPPassScanner.

The Android package name is:
com.nikhil.vippassscanner

The project uses:

- Firebase Authentication
- Cloud Firestore
- Firebase Android SDK

I have cloned the project from GitHub and need to configure my own Firebase project.

Please guide me step-by-step through:

1. Creating a Firebase project
2. Adding the Android app with package name com.nikhil.vippassscanner
3. Downloading and placing google-services.json
4. Enabling Email/Password Firebase Authentication
5. Creating a Cloud Firestore database
6. Creating the passes collection
7. Understanding the required fields
8. Understanding the scanHistory collection
9. Creating appropriate Firestore security rules
10. Testing the application with a sample VIP pass

The expected Firestore structure is:

```
passes/{passCode}
```

- used: boolean
- scannedBy: string
- scannedAt: timestamp

scanHistory/{autoGeneratedId}

- passCode: string
- status: string
- scannedBy: string
- scannedAt: timestamp

The QR/barcode value must match the document ID in the passes collection.

Do not assume I already know Firebase. Give me the steps in order and explain what I need to do in the Firebase

11. Quick Reference

Android package:

com.nikhil.vippassscanner

Firebase services:

- Authentication (Email/Password)
- Cloud Firestore

Firestore collections:

- passes
- scanHistory

Initial pass:

passes/VIP001

used = false

QR/barcode:

VIP001

Real Firebase config:

app/google-services.json

GitHub template:

app/google-services.json.example